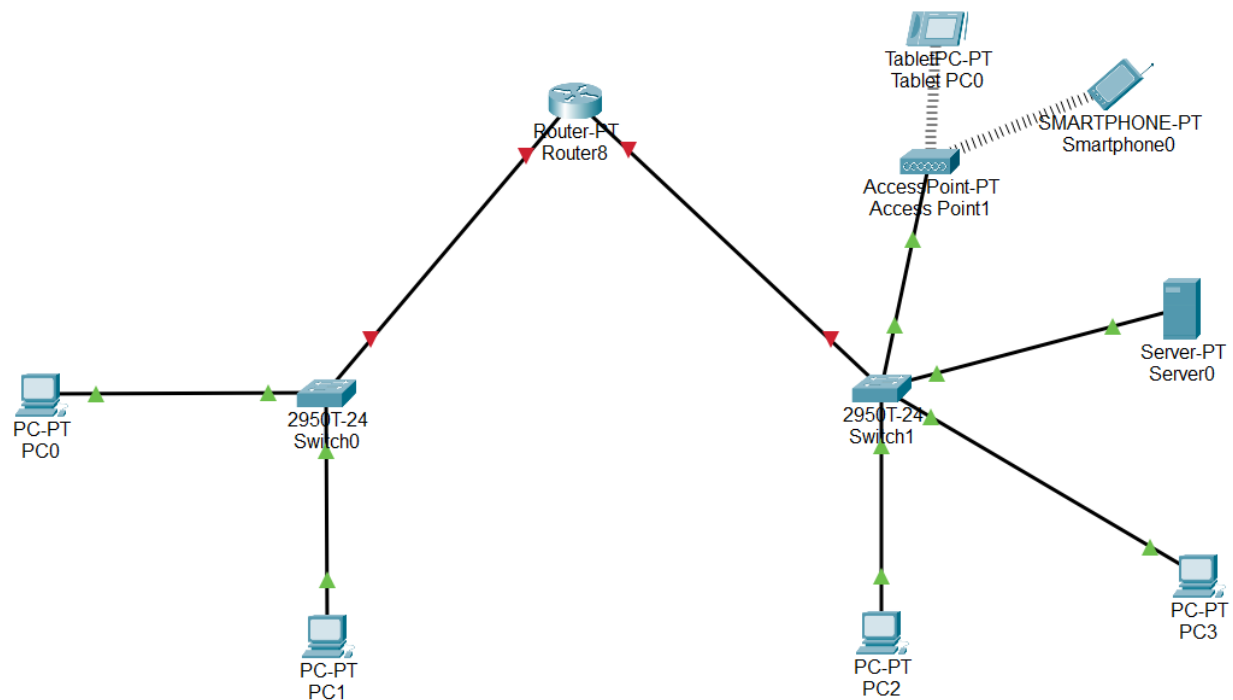


CNSA111 – Fun with Packet Tracer # 2

The goal of this lab is to configure IP addresses both statically and dynamically allowing all devices on both networks to ping.

1. Begin with your Packet Tracer file from Fun with Packet Tracer # 1.
2. Add an AccessPoint-PT device to the network. Connect the AccessPoint-PT device to the same switch your Server-PT is connected to with a copper straight-through cable.
3. Add a TabletPC-PT device.
4. Add a Smartphone-PT.

Your network should look something like this:



5. Access your router configuration information (click the router, select the Config tab). Assign the following IP addresses:
 - a. Router interface connected to Switch0: 192.168.1.1 and subnet mask 255.255.255.0
 - b. Router interface connected to Switch1: 200.2.2.1 and subnet mask 255.255.255.0
 - c. Make sure both router interfaces are turned on.
6. Manually assign the following IP addresses to PC0 and PC1: (click the PC, go to the Config tab, select the NIC interface):
 - a. PC0: 192.168.1.2, subnet mask 255.255.255.0
 - b. PC1: 192.168.1.3, subnet mask 255.255.255.0

7. On the other side of the network infrastructure, set PC2 and PC3 to get IP addresses dynamically through DHCP.
8. Set the Server-PT IP address information to 200.2.2.2 and subnet mask 255.255.255.0.
9. At this point, all arrows should be green indicating active connections. But we still need to set up DHCP on the Server-PT so it can hand out IP addresses to PC2 and PC3.
10. Open the Server-PT properties and select the Services tab. On the left side, select DHCP. Make sure you have the right interface selected at the top of the DHCP screen. Set your default gateway to 200.2.2.1. Set the start IP address to 200.2.2.0 with subnet mask of 255.255.255.0. Be sure to save your modifications.
11. Your DHCP configuration should look like this:

The screenshot shows the 'Server0' configuration window with the 'Services' tab selected. On the left, a list of services includes HTTP, DHCP (selected), DHCPv6, TFTP, DNS, SYSLOG, AAA, NTP, EMAIL, FTP, IoT, VM Management, and Radius EAP. The main area is titled 'DHCP' and shows the configuration for the 'GigabitEthernet1' interface. The 'Service' is set to 'On'. The 'Pool Name' is 'serverPool'. The 'Default Gateway' is '200.2.2.1'. The 'DNS Server' is '0.0.0.0'. The 'Start IP Address' is '200.2.2.0' and the 'Subnet Mask' is '255.255.255.0'. The 'Maximum Number of Users' is '256'. The 'TFTP Server' and 'WLC Address' are both '0.0.0.0'. At the bottom, there are 'Add', 'Save', and 'Remove' buttons. Below these buttons is a table showing the configuration for the 'serverPool'.

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
serverPool	200.2.2.1	0.0.0.0	200.2.2.0	255.255.255.0	256	0.0.0.0	0.0.0.0

12. Close the Server-PT properties.
13. Open the properties of PC2 (or the first PC connected to the same switch as the server). Select the Desktop tab and open a Command Prompt. View your IP address information. If your PC

does not have an IP address at this point use the **ipconfig /renew** command to make it ask the DHCP server for an address.

14. Did the PC get an IP address? **If yes, insert a snip of the address information here.** If not, something went pear-shaped and we need to troubleshoot. What would you look at to begin troubleshooting this issue?

```
C:\>ipconfig

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::202:16FF:FEE3:23ED
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 200.2.2.3
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: ::
                                200.2.2.1

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: ::
                                0.0.0.0
```

15. If the first PC successfully receives an IP address from the DHCP server, repeat the process so the second PC receives an IP address. **Insert snips from all four PCs showing the results of the ipconfig command:**

```
C:\>ipconfig

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::2D0:FFFF:FE91:E36C
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 192.168.1.2
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: ::
                                0.0.0.0

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: ::
                                0.0.0.0
```

```
C:\>ipconfig

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::2D0:BCFF:FE98:93CD
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 192.168.1.3
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: ::
                                0.0.0.0

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: ::
                                0.0.0.0
```

```
C:\>ipconfig

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::202:16FF:FEE3:23ED
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 200.2.2.3
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: ::
                                200.2.2.1

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: ::
                                0.0.0.0
```

```
C:\>ipconfig

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::2D0:58FF:FE5E:1245
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 200.2.2.5
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: ::
                                200.2.2.1

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: ::
                                0.0.0.0
```

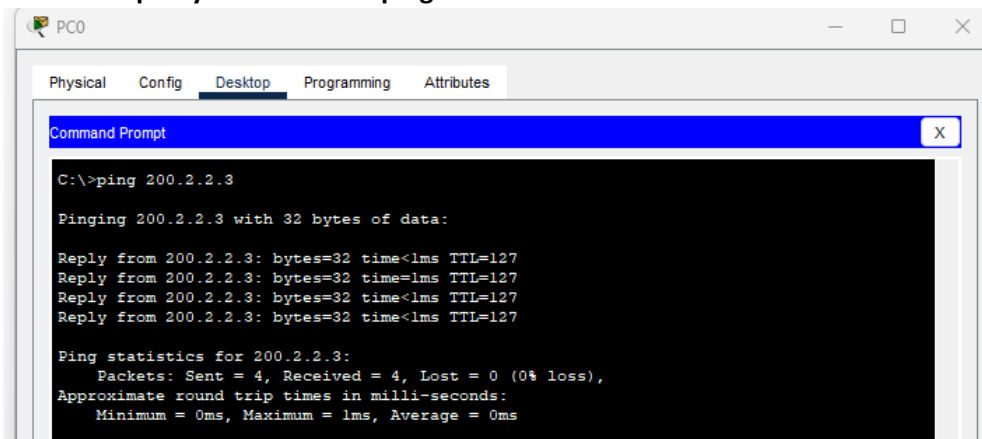
16. Open the Command Prompt utility on the Desktop of PC0 (this is the PC with the statically assigned address 192.168.1.2). Ping PC3 – remember, PC3 got an address from the DHCP server. **Is the ping successful? If not, why not?**

No because the request timed out

17. Resolve the error discovered above. Check all other devices for the same issue.

The reason it didn't work before was because pc0 and pc1 did not have default gateways set so I had to set their default gateways to 200.2.2.1 to match the gateway of devices connected to the server.

18. Attempt to ping from PC0 to PC3 again. Were you successful? If not, troubleshoot connectivity. **Insert a snip of your successful ping here:**



```
PC0
Physical Config Desktop Programming Attributes
Command Prompt
C:\>ping 200.2.2.3

Pinging 200.2.2.3 with 32 bytes of data:

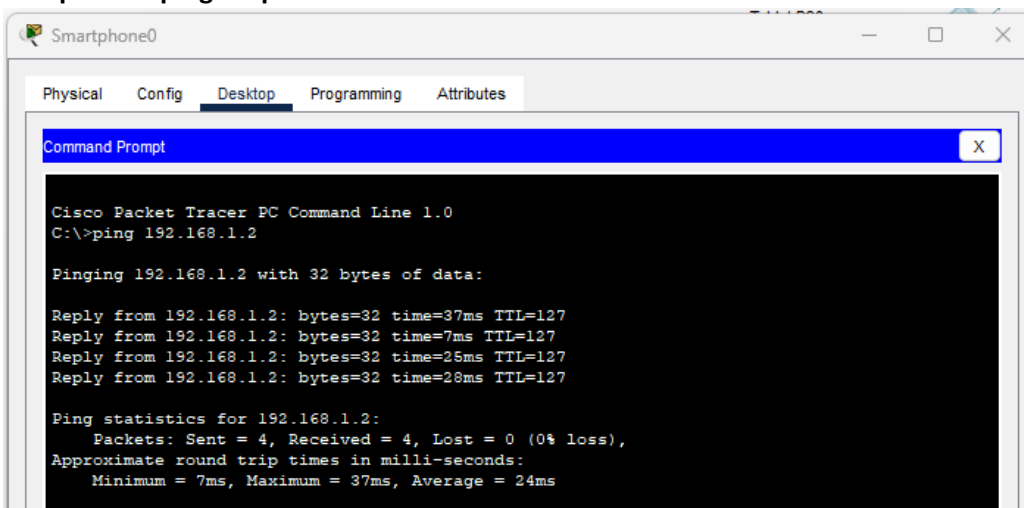
Reply from 200.2.2.3: bytes=32 time<1ms TTL=127
Reply from 200.2.2.3: bytes=32 time<1ms TTL=127
Reply from 200.2.2.3: bytes=32 time<1ms TTL=127
Reply from 200.2.2.3: bytes=32 time<1ms TTL=127

Ping statistics for 200.2.2.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

19. Return to your Access Point configuration. Select the port associated with your Wi-Fi channel and set the SSID to MyNetwork. Select the WPA2-PSK authentication method and set the PSK Pass Phrase to cnsacnsa1.

20. Access the configuration information for your smartphone (Wireless 0). Set the SSID to MyNetwork and authentication to WPA2-PSK. Enter the WAP passphrase.

21. Open a command prompt on the smartphone and try to ping PC0. Is the ping successful? **Insert a snip of the ping output here:**



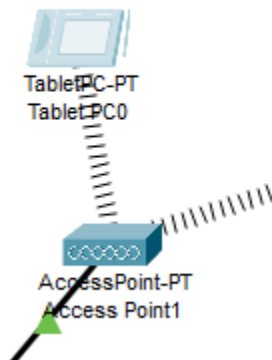
```
Smartphone0
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time=37ms TTL=127
Reply from 192.168.1.2: bytes=32 time=7ms TTL=127
Reply from 192.168.1.2: bytes=32 time=25ms TTL=127
Reply from 192.168.1.2: bytes=32 time=28ms TTL=127

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 7ms, Maximum = 37ms, Average = 24ms
```

22. Configure your tablet with the appropriate SSID and authentication method and check its connectivity. **Include documentation of successful connection here:**



23. Submit your completed lab document and your saved Packet Tracer network to the Canvas assignment.